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Toxicant default guideline values for aquatic ecosystem protection

Chromium (III) in freshwater

Technical brief

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Summary

The default guideline values (DGVs) and associated information in this technical brief should be used in accordance with the detailed guidance provided in the *Australian and New Zealand Guidelines for Fresh and Marine Water Quality* website (ANZG 2018).

In natural freshwater, chromium (III) is present in low concentrations due to its low solubility (5 µg/L). However, the toxicity of chromium (III) is known to be associated with dissolved, colloidal and precipitated forms. Chromium (III) readily forms stable complexes with both organic and inorganic ligands and can also be strongly adsorbed. Factors affecting chromium (III) speciation include pH, temperature, organic complexation, alkalinity, water hardness and redox conditions. Manganese oxides, a dominant oxidiser in aquatic systems, are responsible for the oxidation of chromium (III) to the more toxic chromium (VI). Chromium (VI) can also be reduced by sulfide ions, iron (II), fulvic acids, low molecular-weight organic compounds and microbial activity.

The revision of these DGVs highlighted a lack of toxicity data for chromium (III), especially chronic and high-quality data – only 3 studies reported measured chromium test concentrations (dissolved or total) and the minimum necessary supporting water-quality parameters. The low aqueous solubility of chromium (III) resulted in the toxicity observed being due to a combination of dissolved, colloidal and precipitated chromium (III) in most of the toxicity tests.

Thirteen species from 5 taxonomic groups (one protozoan, 3 microalgae, 2 microcrustaceans, 2 rotifers, and 5 fish) from toxicity tests with a pH of 6.0–8.3, water hardness of < 100 mg/L $CaCO_3$ and alkalinity of 18–42 mg/L $CaCO_3$ (when reported) passed the quality-assessment and screening processes and were used to derive the DGVs. The data comprised 6 chronic toxicity values (chronic NOEC and EC16 data, and chronic EC50 data converted to EC10 values) and 7 acute toxicity values, which were converted to estimated chronic no-effect or low-effect values where necessary. The fit of the species sensitivity distribution to the data was good, resulting in moderate reliability DGVs. The chromium (III) DGVs for 99%, 95%, 90% and 80% species protection are 0.95 µg/L, 6.7 µg/L, 16 µg/L and 39 µg/L, respectively. The 95% species-protection level for total chromium (III) in freshwater (6.7 µg/L) is recommended when assessing ecosystems that are slightly to moderately disturbed. However, the accuracy of the DGVs may be affected due to uncertainties in deriving DGVs for metals of low solubility, the lack of high-quality toxicity studies, the predominance of chronic EC50 data and acute data that needed to be converted to negligible-effect estimates, and the wide range of factors influencing chromium (III) speciation. When comparing chromium (III) concentrations in water samples to the DGVs, the chromium (III) concentration should be based on either total chromium (III) or, preferably, the potentially bioavailable fraction (i.e. pH-1.75 extractable) of chromium (III).

Given the wide range of water-quality parameters that affect the speciation and oxidation state of chromium and the endemic nature of chromium (III) in environmental systems, site-specific guideline values and background concentrations should also be considered. This is especially so where the pH, hardness or alkalinity of a waterbody is outside the range relevant to the toxicity data upon which the DGVs were derived. More chronic toxicity data for a broader range of species and taxonomic groups and, ideally, assessing the effects of key toxicity-modifying factors are needed to derive DGVs with higher reliability.

1 Introduction

Chromium is a metal that can exist in several different oxidation states (−II to +VI). The most common states in the aquatic environment are chromium (III) and chromium (VI). Naturally occurring chromium is nearly always present as chromium (III). In surface water, the dominant oxidation state is the hexavalent (VI) form, while chromium (III) (the most stable oxidation state) dominates in mildly reducing environments, such as sediments and wetlands (Bailar et al. 1973). It is unclear if chromium (III) is an essential nutrient for aquatic freshwater plants and animals. Historically, it has been reported to be responsible for the control of glucose and lipid metabolism in animals and was considered an essential trace element for mammals and some microorganisms (WHO 2009); however, this may not necessarily be correct (Vincent 2010).

Chromium from human activity enters the environment through emissions from metallurgy and metal finishing, pigments, paints, leather tanning, wood preservation and corrosion inhibitors, and through its use as a chemical intermediate or catalyst. Smaller quantities of chromium are used in cosmetics, toners for copying machines, magnetic tapes, fertilisers, rubber products, plastics, soaps and cleaning products. Natural sources of chromium include volcanic emissions, forest fires, vegetative debris and marine aerosols. Surface runoff, deposition from air, and municipal and industrial wastewater are the most common sources of chromium (III) and chromium (VI) in surface water (Pawlisz et al. 1997; WHO 2009).

The speciation of chromium (III) in surface water is dependent on pH, temperature, organic complexation (especially ligands containing oxygen, nitrogen or sulfur), alkalinity, water hardness and redox conditions. In the absence of complexing ligands and under anaerobic conditions, the main dissolved phases of chromium (III) are Cr^{3+} (pH < 4), $CrOH^{2+}$ (pH 4–6.5) and $Cr(OH)_4^-$ (pH > 11.5). At the pH of natural waters (6.5–9), sparingly soluble chromium ($Cr(OH)_3$) dominates (Rai et al. 1987), and its solubility is estimated as < 5 µg/L (Rai et al. 1989).

Chromium (III) readily forms stable complexes with both organic and inorganic ligands, such as hydroxyl, sulfate, ammonium, cyanide, sulfocyanide, fluoride, chloride, and natural and synthetic organic compounds. Chromium (III) complexes can also be strongly adsorbed to natural particulates, which contributes to the decrease in chromium (III) mobility and bioavailability (Mattuck and Nikolaidis 1996). Manganese oxides are the most significant oxidants in the environment responsible for oxidising chromium (III) to chromium (VI), but a large excess is required (Vignati et al. 2010). Oxygen is of negligible influence (Saleh et al. 1989). Chromium (VI) is a potential source of chromium (III) following reduction by sulfide ion, iron (II), fulvic acid, low molecular-weight organic compounds and microbial activity (Kotaś and Stasicka 2000). Unlike chromium (III), chromium (VI) readily crosses cell membranes and, once inside the cells, is readily reduced to chromium (III). However, Aharchaou et al. (2022) reported that chromium (III) was similarly bioavailable as chromium (VI) during acute (96-hour) exposures to the freshwater mussel *Corbicula fluminea*.

Natural concentrations of chromium in surface water are approximately 0.026–10 µg/L (WHO 1988; Kotaś and Stasicka 2000), and dissolved chromium concentrations of < 0.1 µg/L are reported when using trace-metal analysis techniques (Shiller and Boyle 1991). An uncontaminated river in Poland contained total chromium (III) concentrations of 0.52 µg/L (Bobrowski et al. 2004). Concentrations of

chromium (III) as high as 104 µg/L have been reported in industrial areas (Domínguez Renedo et al. 2004), especially near tanneries (WHO 1988). Wastewater from industry (Tang et al. 2004; Ding et al. 2005), electroplating and dye plants (Prasada et al. 1998; Kiptoo et al. 2004, Hashemi et al. 2004) ranged from 5 µg/L to 4,000 µg/L chromium (III), and values from tannery effluents were even higher (Prasada et al. 1998; Domínguez and Arcos 2002; Domínguez Renedo et al. 2004).

Standard methods for the analysis of chromium continue to be challenging, as they are susceptible to contamination. Online separation and detection methods, such as high-performance liquid chromatography inductively coupled plasma mass spectrometry (HPLC-ICP-MS), are proving to be promising but they are challenging for complex wastewater streams, such as tannery effluents rich in organic compounds, and they are costly (Šĉanĉar and Milaĉiĉ 2014; Jabłońska-Czapla 2015). Common analytical techniques include ultraviolet-visible spectrometry (UV-Vis), atomic-absorption spectrometry (AAS), electrothermal atomic-absorption spectrometry (ETAAS) and ICP atomic emission spectrometry (ICP-AES) coupled with on-line or off-line separation and preconcentration of a specific chromium species; however, potential issues associated with matrix, sampling, preservation and storage may exist (Kotaś and Stasicka 2000). Most methods require separation of chromium (III) via precipitation, and its determination is the difference in total dissolved chromium before and after precipitation (e.g. Frois et al. 2012, Procházková et al. 2019 and references therein).

Given the substantial differences in the bioavailability and toxicity of chromium (III) compared to chromium (VI), there is sufficient justification to derive default guideline values (DGVs) for each oxidation state. The natural widespread presence of chromium (III) in the environment and the multiple factors influencing its bioavailability and toxicity (Adams et al. 2000) in freshwater indicate that site-specific guideline values should also be considered.

The ANZECC and ARMCANZ (2000) DGV for chromium (III) was 3.3 µg/L, calculated by applying an assessment factor of 20 to the lowest available toxicity value – an LC50 value (see 'Glossary and acronyms' for definitions) for the rainbow trout (*Oncorhynchus mykiss*). The DGV was considered to be an indicative interim working level only, given the uncertainty associated with it. In the current DGV derivation method (Warne et al. 2018), the use of assessment factors is not recommended, and values derived in this way are considered of unknown reliability. Water-quality guideline values for chromium (III) in freshwater in the United Kingdom (Maycock et al. 2007) and Canada (CCME 1999) are 4.7 µg/L (or 32 µg/L for short-term exposure) and 8.9 µg/L (interim guideline value), respectively, and were based on applying an assessment factor of 10 to the most sensitive species (and endpoint) tested.

2 Aquatic toxicology

Summaries of the acceptable quality raw chronic and acute chromium (III) toxicity data for all freshwater species that passed the quality-assessment process are provided in Section 2.1 and Section 2.2, respectively.

2.1 Chronic toxicity

A 48-hour EC50 of 95 µg/L chromium (III) at a water hardness of 2.8 mg/L $CaCO_3$ was reported by Nalecz-Jawecki and Sawicki (1998) for the protozoan *Spirostomum ambiguum*. The single-celled microalga *Raphidocelis subcapitata* (formerly known as *Pseudokirchneriella subcapitata*) was also highly sensitive. Vignati et al. (2010) reported a 72-hour EC50 of 17 µg/L for growth inhibition of *R. subcapitata* (calculated as a time-weighted mean [TWM] using measured chromium concentrations in the medium used in the ISO 8692:2012 standard), while a 72-hour EC50 of 105 µg/L in Algal Assay Procedure medium (AAP; unmeasured chromium concentrations) was reported by Hickey et al. (1991). A more recent study by Alho et al. (2022) found *R. subcapitata* to be less sensitive to chromium (III) than reported by Vignati et al. (2010), reporting a significant stimulatory effect on growth rate at 200 µg/L and a significant reduction in growth rate at 850 µg/L. The microalga *Chlorella kessleri* was also sensitive to chromium (TWM 72-hour EC50 of 50 µg/L; Vignati et al. 2010). *Dictyosphaerium chlorelloides* was less sensitive than other microalga to chromium (72-hour EC50 of 2,786 µg/L; Pereira et al. 2013). Biesinger and Christensen (1972) observed greater sensitivity of the cladoceran *Daphnia magna* to chromium in soft water (45 mg/L $CaCO_3$), reporting a 21-day EC16 of 330 µg/L and EC50 of 600 µg/L compared to the 21-day NOEC of 700 µg/L in hard water (350 mg/L $CaCO_3$) reported by Kühn et al. (1989). Biesinger and Christensen (1972) also observed mortality as an endpoint in the 21-day exposures, reporting an LC50 of 2,000 mg/L in soft water. A study with *D. magna* that measured concentrations of chromium, water hardness (269 mg/L $CaCO_3$) and alkalinity (43 mg/L $CaCO_3$) reported a 21-day reproductive TWM EC50 of 1,900 µg/L (Ponti et al. 2014). A fish species, *O. mykiss*, was sensitive to chromium – early life-stage development tests measuring hatching (32-day) and development of feeding fry (72-day) from fertilised eggs generated NOEC values of 89 ± 15 µg/L and 48 ± 3 µg/L, respectively (25 mg/L $CaCO_3$). Despite pseudo-replication, these data were considered to be of acceptable quality because chromium concentrations and water-quality parameters were measured throughout the toxicity test (at least weekly) in non-static (flow-through) exposures (Stevens and Chapman 1984). Although failing the quality-assessment criteria, chronic toxicity to the insect *Acroneuria lycorias* was studied by Warnick and Bell (1969), who reported a 7-day EC50 of 32,000 µg/L, indicating this species (and potentially this taxon) was not sensitive to chromium.

2.2 Acute toxicity

Hernández-Ruiz et al. (2016) reported 24-hour LC50 values of approximately 1,050 µg/L and 640 µg/L for 2 strains of the rotifer *Brachionus calyciflorus*. The rotifer *Lecane quadridentata* was slightly less sensitive (24-hour LC50 of 1,279 µg/L) (Hernández-Ruiz et al. 2016). The cladoceran *D. magna* had a 24-hour EC50 of 22,000 µg/L in hard water (350 mg/L $CaCO_3$) (Kühn et al. 1989) and a measured TWM 48-hour EC50 of > 5,300 µg/L and 14,100 µg/L (depending on water alkalinity) (Ponti et al. 2014). *Daphnia similis* was more sensitive than *D. magna*, (48-hour EC50 of 3,240 µg/L and NOEC of

2,500 µg/L in soft water [44 mg/L $CaCO_3$]) at a lower pH of 6.0–6.5, which was used to increase the proportion of dissolved chromium (III) in toxicity test treatments (Melnikov and de Freitas 2011). Acute toxicity values observed for the fish *O. mykiss* were lower than chronic toxicity values (96-hour LC50 of 4,400 µg/L and NOEC of 3,100 µg/L based on measured chromium concentrations; Stevens and Chapman 1984). Toxicity tests measuring the mortality of an additional 6 fish species generated 96-hour LC50 values ranging from 3,300 µg/L to 7,460 µg/L in soft water (20 mg/L $CaCO_3$) (Pickering and Henderson 1966) and from 57,000 µg/L to 144,000 µg/L in hard water (300 mg/L $CaCO_3$ and 360 mg/L $CaCO_3$) (Pickering and Henderson 1966; Azmat and Javed 2011; Venkatachalam and Natarajan 2014).These studies showed that younger fish (60-day compared to 90-day and 120-day) and longer exposure durations (96 hours compared to 24 hours and 48 hours) were the most sensitive. Acute toxicity data for the insect *Ephemerella subvaria* failed the quality-assessment criteria. However, a 96-hour LC50 of 2,000 µg/L suggested that this species (and potentially this taxon) is more acutely sensitive than fish and is of comparable sensitivity to cladocerans (Warnick and Bell 1969). Data for which the quality could not be assessed but were expected to be from an acceptable study design included the 96-hour LC50 of approximately 100 µg/L (extrapolated from a figure) for *Daphnia pulex* (Stackhouse and Benson 1989).

3 Factors affecting toxicity

A range of factors influence the speciation, bioavailability and toxicity of chromium (III) to freshwater biota, including pH, water hardness, alkalinity, redox potential, presence of organic (e.g. dissolved organic carbon [DOC]) and inorganic ligands. Oxidants (e.g. manganese oxides) and reductants (e.g. sulfide ion, iron) also influence the oxidation state of chromium in natural waters; hence, the more toxic chromium (VI) should also be considered. Due to the low solubility of chromium (III), dissolved, polymeric colloidal and precipitated phases are all significant sources of exposure for freshwater biota. These reactions are seen as being more important than oxidation, resulting in losses of dissolved chromium (III) with time (Vignati et al. 2010).

The solubility of chromium (III) increases with decreasing pH, ultimately influencing toxicity to biota. However, toxicity to the amphipod *Hyalella azteca* (28-day and 42-day survival) increased with increasing pH (decreasing measured, filterable chromium (III)), suggesting that toxicity involves more than a pH-dependent difference in solubility (Besser et al. 2004). A nominal concentration of 100 µg/L resulted in measured filterable concentrations of 29 µg/L, 48 µg/L and 76 µg/L at pH of 7.9, 7.2 and 6.4, respectively. In contrast, reproduction of *H. azteca* was not affected at a pH of 7.2 and was substantially higher in the chromium (III) treatment compared to the control treatment at pH 6.4 and 7.9. Particulate (including colloidal) chromium (III) is likely to be a significant source of chromium (III) exposure to biota. Histopathology studies also indicate a high deposition of chromium (III) on fish gills (Wong et al. 1982).

An increase in water hardness has been shown to decrease the toxicity of chromium (III) (refer to Section 2.2). US EPA (1986) derived a hardness algorithm to account for this variation and to enable direct comparison of toxicity values at different water-hardness values, and this approach was adopted in ANZECC and ARMCANZ (2000). However, the validity of the hardness algorithm is questionable because:

- it was derived using a limited dataset (one cladoceran and 2 fish species) at only 2 water hardness values (low and high)
- it included the use of unpublished data (cladoceran)
- it did not differentiate between dissolved and particulate exposure pathways.

Hence, further studies are required to quantify the effects of water hardness and other key toxicity-modifying factors on chromium toxicity and associated guideline values. Therefore, the hardness correction for chromium (III) toxicity values and DGVs is no longer recommended for use in Australia and New Zealand. This is consistent with the recommended application of chromium (III) guideline values in the United Kingdom (Maycock et al. 2007) and Canada (CCME 1999).

Alkalinity has also been shown to influence the toxicity of chromium (III) (Ponti et al. 2014). Studies with the cladoceran *D. magna* used measured dissolved concentrations of chromium throughout toxicity tests with toxicity values (LC50s) based on both nominal and measured TWMs. Test solutions at 2 alkalinities (245 mg/L $CaCO_3$ and 39 mg/L $CaCO_3$) and hardness (240–250 mg/L $CaCO_3$) showed that the formation of insoluble chromium (III) oxyhydroxides was rapid (within 6 hours) in high-alkalinity water and slow in low-alkalinity water (22% to 43% decrease in soluble chromium after 48 hours). This was reflected in the difference in toxicity to *D. magna* – LC50s of > 80 mg/L and

18.3 mg/L (nominal chromium concentrations) were observed for high-alkalinity water and low-alkalinity water, respectively. The lack of toxicity in high-alkalinity water at a nominal concentration of up to 80 mg/L compared to almost-complete (90%) effect at only 20 mg/L (nominal) in low-alkalinity water suggests that exposure to precipitated insoluble chromium (III) oxyhydroxides does not cause toxicity (mortality) to *D. magna* (Ponti et al. 2014).

Increasing DOC can decrease the toxicity of chromium (III). The 72-hour TWM EC50 value for the microalga *P. subcapitata* increased from 17 ± 5 µg/L in ISO medium with 0.4 mg/L DOC to 245 ± 47 µg/L in ISO medium prepared using ultrafiltered lake water with 1.4 mg/L DOC (Vignati et al. 2010). However, the change in toxicity was also likely due to hardness and the formation of hydrolysis products (e.g. chromium hydroxides) (Vignati et al. 2010). The addition of humic acid from 0 mg/L to 50 mg/L decreased acute toxicity of chromium (III) to *Daphnia pulex* by a factor of 60, and as little as 0.5 mg/L humic acid decreased toxicity by about a factor of 2 (Stackhouse and Benson 1989).

Measured concentrations of metals in toxicity test media and throughout the duration of toxicity tests are crucial when interpreting toxicity data. Initial concentrations of dissolved chromium (III) in an algal toxicity test were found to decrease by 20% to 30% compared to nominal concentrations, and further decreases of 60% to 90% occurred after 72 hours (the largest decrease occurred within the first 24 hours) (Vignati et al. 2008, 2010); hence non-static exposures or calculation of TWM toxicity values was used in some studies to attempt to overcome this issue (Stevens and Chapman 1984; Vignati et al. 2010; Ponti et al. 2014). The low solubility of some metals poses a unique challenge to deriving reliable toxicity values including, but not limited to, considerations of:

- metal binding to organism surfaces
- the rate of loss of dissolved metals from the test solution
- sub-sampling for metal analyses (dissolved, colloidal, precipitate)
- the need for renewal (e.g. ideally after 24 hours) of test solutions with fresh test solutions or, preferably, the use of flow-through, continuous renewal (e.g. Stevens and Chapman 1984).

Given the range of factors affecting the toxicity of chromium (III) in freshwater, detailed characterisation of test waters and measured chromium concentrations is recommended, as these parameters significantly influence its speciation, bioavailability and toxicity to freshwater biota. Due to the lack of studies that reported this information, further studies are required to quantify the effect of these parameters on toxicity and to derive DGVs of high reliability.

4 Default guideline value derivation

The DGVs were derived in accordance with the method described in Warne et al. (2018), except that the shinyssdtools software (V 0.4.0) (Dalgarno 2018) was used instead of the Burrlioz 2.0 software.

4.1 Toxicity data used in derivation

Only studies that used chromium (III) salts as the test compound were assessed for quality and acceptability. Toxicity values for 16 species from 5 taxonomic groups (protozoan, microalga, rotifer, microcrustacean and fish) passed the quality-assessment process. The data consisted of a chronic NOEC and a chronic EC16 value, estimated chronic no-effect or low-effect values from chronic EC50 values, and estimated chronic no-effect or low-effect values from acute LC50 values. The chronic EC16 reported for the cladoceran *D. magna* was considered to be an acceptable no-effect or low-effect toxicity value (i.e. no conversion factor was applied to this value), as it was the lowest toxicity value for this species and there were no lower-effect toxicity measures (e.g. EC10) available.

The available toxicity data were generated using chloride, nitrate and potassium sulfate salts of chromium (III). Due to the low solubility of chromium (III) at a neutral pH, both dissolved and precipitated phases were considered collectively as major sources of exposure for freshwater biota. The solubility of chromium (III) salts used to generate toxicity data is generally high (Aylward and Findlay 1994); however, the resulting pH of these solutions is low – a pH of less than 4.6 was reported for concentrations of ≥ 64 mg/L (Warnick and Bell 1969; Hale 1977; Martin and Holdich 1986). For these studies, the low pH was identified as a confounding factor when interpreting the toxicity of chromium (III); hence, to be acceptable, the pH of toxicity test solutions needed to be measured and confirmed to be within the range of natural waters. Unfortunately, DOC, which is another important toxicity-modifying factor for chromium (III), was not reported for any of the data that were used to derive the DGVs. Only 3 studies, comprising toxicity data for 4 species, reported toxicity values based on measured concentrations of total or dissolved or colloidal chromium (III): 2 microalgae (Vignati et al. 2010), one cladoceran (Ponti et al. 2014) and one fish (Stevens and Chapman 1984). In these studies, chromium was assumed to be present as chromium (III). All other studies reported nominal concentrations only. This is an important limitation of the dataset, and it is highly desirable that these tests be repeated with test concentrations measured at both the beginning and end of the toxicity tests.

The majority of the toxicity tests from which the toxicity values were derived used water with a hardness of less than 100 mg/L $CaCO_3$; however, there were 3 exceptions: toxicity tests for the 3 fish species *Catla catla*, *Cirrhinus mrigala* and *Labeo rohita*, which were conducted at a very high hardness (~300 mg/L $CaCO_3$) (Azmat and Javed 2011; Venkatachalam and Natarajan 2014). The toxicity of chromium (III) to these species was approximately an order of magnitude lower than for any other species in the dataset – 96-hour LC50s of between 57,000 µg/L and 80,000 µg/L were observed. Although the test water hardness for *L. rohita* was not specified (Venkatachalam and Natarajan 2014), it was concluded as very likely to have been very high hardness given that Azmat and Javed (2011) also found that this species was highly insensitive to chromium (III) at a very high hardness of 300 mg/L $CaCO_3$ – a 96-h LC50 of 86,000 µg/L was observed, which is similar to the 57,000 µg/L value reported by Venkatachalam and Natarajan (2014). Given the effect that high

hardness has on chromium (III) toxicity to fish (also demonstrated by Pickering and Henderson 1966; see Section 2.2), the 3 fish species tested at high hardness were excluded from the final dataset. Thus, the final dataset comprised toxicity data generated at pH 6–8.3, hardness < 100 mg/L $CaCO_3$ and alkalinity 18–42 mg/L $CaCO_3$, where reported. Notwithstanding the body of information on factors that modify chromium (III) toxicity, the quantitative incorporation of such factors into the DGVs was beyond the scope of the current derivation.

A summary of the toxicity data (one value per species) used to calculate the DGVs for total chromium (III) in freshwater is provided in Table 1. Further details of the water-quality parameters for each single-species value used to calculate the DGVs are presented in Appendix A. Details of the data-quality assessment and the data that passed the quality assessment are provided as supporting information. The final dataset comprised chronic toxicity values for 6 species from 4 taxonomic groups and acute toxicity values for 7 species from 3 taxonomic groups (converted to estimated chronic no-effect or low-effect values where necessary, as per Warne et al. 2018), totalling 13 species from 5 taxonomic groups (protozoan, microalga, rotifer, microcrustacean and fish). Each toxicity value was representative of the most sensitive life stage, toxicity test duration and endpoint measured, based on Warne et al. (2018).

An assessment was undertaken based on deriving the DGVs using the 6 chronic toxicity values only or using the dataset of 13 values comprising both the chronic toxicity data and acute toxicity data converted to chronic-equivalent data, excluding the data for the 3 fish species at high hardness (Appendix B). The protective concentrations based on chronic toxicity data only were considered too conservative based on the reported range of natural background concentrations of chromium (III). The combined chronic and acute (converted to chronic) toxicity dataset yielded slightly less conservative protective concentrations and did not exhibit bimodality.

Table 1 Summary of single toxicity values for all species used to derive the default guideline values for total chromium (III) in freshwater

Taxonomic group (phylum)	Species	Life stage	Duration (h)	Type (acute or chronic)	Toxicity measure (test endpoint)	Reported toxicity value (µg/L)	Final toxicity value[a] (µg/L)
Protozoa (Ciliophora)	*Spirostomum ambiguum*	NR	48	Chronic	EC50 (deformities)	95	19
Microalga (Chlorophyta)	*Raphidocelis subcapitata*	Exponential growth	72	Chronic	EC50 (growth)	17	3.4
Microalga (Chlorophyta)	*Chlorella kessleri*	Exponential growth	72	Chronic	EC50 (growth)	50	10
Microalga (Chlorophyta)	*Dictyosphaerium chlorelloides*	Exponential growth	72	Chronic	EC50 (growth)	2,786	557
Rotifer (Rotifera)	*Brachionus calyciflorus*	Neonates	24	Acute	LC50 (mortality)	820[b]	82
Rotifer (Rotifera)	*Lecane quadridentata*	Neonates	24	Acute	LC50 (mortality)	1,279	128
Crustacean (Arthropoda)	*Daphnia magna*	Neonates	504	Chronic	EC16 (reproduction)	330	330
Crustacean (Arthropoda)	*Daphnia similis*	Neonates	48	Acute	EC50 (immobilisation)	3,240	324

Taxonomic group (phylum)	Species	Life stage	Duration (h)	Type (acute or chronic)	Toxicity measure (test endpoint)	Reported toxicity value (µg/L)	Final toxicity value[a] (µg/L)
Fish (Chordata)	*Oncorhynchus mykiss*	Fertilised eggs	1,728	Chronic	NOEC (feeding fry)	48	48
Fish (Chordata)	*Poecilia reticulata*	NR	96	Acute	LC50 (mortality)	3,330	333
Fish (Chordata)	*Carassius auratus*	NR	96	Acute	LC50 (mortality)	4,100	410
Fish (Chordata)	*Pimephales promelas*	NR	96	Acute	LC50 (mortality)	5,070	507
Fish (Chordata)	*Lepomis macrochirus*	NR	96	Acute	LC50 (mortality)	7,460	746

NR = not reported.

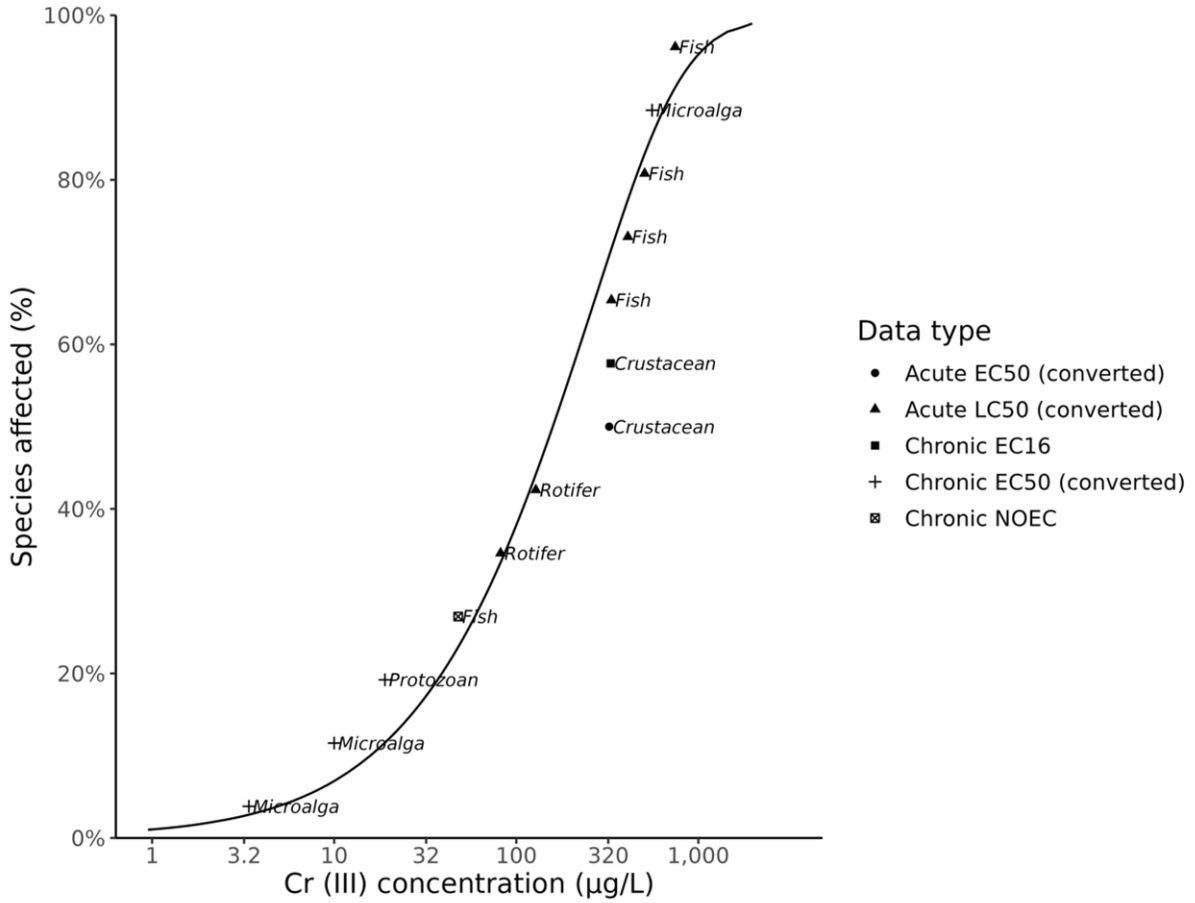
[a] Data conversions to estimate chronic no-effect or low-effect estimates: chronic EC50 values were divided by 5; acute EC50 or LC50 values were divided by 10; chronic NOEC and EC16 – no conversion.

[b] Geometric mean (n = 2).

4.2 Species sensitivity distribution

The cumulative frequency (species sensitivity) distribution (SSD) of the 13 freshwater chronic and acute (converted to chronic) toxicity data for total chromium (III) reported in Table 1 is shown in Figure 1. The SSD was plotted using the shinyssdtools (V 0.4.0) software. The model was judged to provide a good fit to the data, notwithstanding the offset between the 2 crustacean toxicity values and the SSD at around the 50th percentile. Further details on the SSD fitting process are provided in Appendix C.

Figure 1 Species sensitivity distribution for chromium (III) in freshwater



4.3 Default guideline values

It is important that the DGVs (Table 2) and associated information in this technical brief are used in accordance with the detailed guidance provided in the *Australian and New Zealand Guidelines for Fresh and Marine Water Quality* (ANZG 2018).

The DGVs for 99%, 95%, 90% and 80% species protection are shown in Table 2. The 95% species-protection DGV for chromium (III) in freshwater (6.7 µg/L) is recommended when assessing ecosystems that are slightly to moderately disturbed. When applying the DGV for chromium (III) to an unfiltered water sample, it is recommended to distinguish the chromium associated with mineralised forms in suspended sediment, which is not considered bioavailable, from the potentially bioavailable chromium in colloidal or precipitated and surface-adsorbed forms. To do this, an acid (pH-1.75) extraction of the unfiltered water sample is recommended, as detailed by ANZG (2025). The acid-extractable fraction will contain dissolved or colloidal and precipitated forms without a large contribution from mineralised particulate chromium. Once acidified, all chromium in the sample will be present as chromium (III), even if some chromium (VI) was present in the unacidified sample (ANZG 2025). Consequently, if the acid-extractable chromium concentration exceeds the chromium (III) DGV, a separate chromium (VI) analysis may be required to characterise the relative proportions of chromium (III) and chromium (VI) in the original sample. It is also possible to compare the total chromium concentration of the unfiltered water sample with the DGV but if this fraction exceeds the DGV, it will not be possible to identify the relative proportions of non-bioavailable and bioavailable

chromium in the sample. Further details on applying the DGVs to metals of low solubility, including chromium (III), are presented in Appendix D.

Table 2 Toxicant default guideline values for total chromium (III) in freshwater with moderate reliability

Level of species protection (%)	Default guideline value (µg/L)[a,b]
99	0.95
95	6.7
90	16
80	39

[a] The default guideline values were derived using the shinyssdtools (V 0.4.0) software and have been rounded to 2 significant figures. The accuracy of the default guideline value may be affected due to uncertainties of deriving default guideline values for metals of low solubility, the lack of high-quality toxicity studies, and the wide range of factors influencing chromium (III) speciation. See Section 3 and Appendix D for further details.
[b] The total chromium (III) concentration or, preferably, the potentially bioavailable fraction (i.e. pH-1.75 extractable) of the unfiltered environmental water sample should be compared with the default guideline values. Filtered/dissolved fractions of chromium (III) should not be compared with the default guideline values. See Section 4.3 and Appendix D for further details.

Note that the ANZECC and ARMCANZ (2000) hardness correction for chromium (III) should not be used in conjunction with the DGVs presented here (see Section 3). The DGVs are based on toxicity data generated at pH 6.0–8.3, hardness < 100 mg/L $CaCO_3$ and alkalinity 18–42 mg/L $CaCO_3$, where reported. Where local water quality is outside the range of these variables, site-specific guideline values may need to be considered.

4.4 Reliability classification

The chromium (III) freshwater DGVs have a moderate reliability classification (Warne et al. 2018) based on the outcomes for the following 3 criteria:

- sample size – 13 (good)
- type of toxicity data – chronic and converted acute data
- SSD model fit – good.

Notwithstanding the moderate reliability classification of the DGVs, the accuracy of the DGVs may be affected due to uncertainties in deriving DGVs for metals of low solubility, the lack of high-quality toxicity studies, the predominance of chronic EC50 data and acute data that needed to be converted to negligible effect estimates, and the wide range of factors influencing chromium (III) speciation. More data on the chronic toxicity of chromium (III) to a broader range of freshwater species and taxonomic groups (based on measured chromium (III) concentrations) are needed in order to derive DGVs with higher reliability. Additionally, more research on the effects of toxicity-modifying factors, such as pH, hardness, alkalinity and DOC, on chromium (III) toxicity would help enable the development of bioavailability-based DGVs for chromium (III) in freshwater.

Glossary and acronyms

Term	Definition
AAS	Atomic-absorption spectrometry
Acute toxicity	A lethal or adverse sub-lethal effect that occurs as a result of a short (relative to the organism's life span) exposure to a chemical. Refer to Warne et al. (2018) for examples of acute exposures
Assessment factor (AF)	A unitless number applied to the lowest toxicity figure for a chemical to derive a concentration that should not cause adverse environmental effects. The size of the AF varies with the type of data. Also called 'application factor' or 'safety factor'
CAS no. (Chemical Abstracts Service number)	Each chemical has a unique identifying number that is allocated to it by the American Chemical Society
Chronic toxicity	A lethal or sub-lethal adverse effect that occurs as the result of exposure to a chemical for a period that is a substantial portion of the organism's life span or an adverse sub-lethal effect on a sensitive early life stage. Refer to Warne et al. (2018) for examples of chronic exposures
Default guideline value (DGV)	A guideline value recommended for generic application in the absence of a more specific guideline value (e.g. a site-specific value) in the *Australian and New Zealand Guidelines for Fresh and Marine Water Quality*. Formerly known as 'trigger values'
DOC	Dissolved organic carbon
ECx	The concentration of a substance in water or sediment that is estimated to produce an x% change in the response being measured or a certain effect in x% of the test organisms, under specified conditions
Endpoint	The specific response of an organism that is measured in a toxicity test (e.g. mortality, growth, a particular biomarker)
ETAAS	Electrothermal atomic-absorption spectrometry
Guideline value	A measurable quantity (e.g. concentration) or condition of an indicator for a specific community value below which (or above which, in the case of stressors such as pH, dissolved oxygen and many biodiversity responses) there is considered to be a low risk of unacceptable effects occurring to that community value. Guideline values for more than one indicator should be used simultaneously in a multiple-lines-of-evidence approach. (Also refer to default guideline value and site-specific guideline value)
HPLC-ICP-MS	High-performance liquid chromatography inductively coupled plasma mass spectrometry
ICP-AES	Inductively coupled plasma atomic emission spectrometry
LCx	The concentration of a substance in water or sediment that is estimated to be lethal to x% of a group of test organisms under specified conditions
Lowest-observed-effect concentration (LOEC)	The lowest concentration of a material used in a toxicity test that has a statistically significant ($p \leq 0.05$) adverse effect on the exposed population of test organisms as compared with the controls. All higher concentrations should also cause statistically significant effects
No-observed-effect-concentration (NOEC)	The highest concentration of a toxicant used in a toxicity test that does not have a statistically significant ($p \leq 0.05$) adverse effect on the exposed population of test organisms as compared with the controls
PC	Protective concentration
Site-specific guideline value	A guideline value that is relevant to the specific location or conditions that are the focus of a given assessment or issue

Term	Definition
Species sensitivity distribution (SSD)	A method that plots the cumulative frequency of species' sensitivities to a toxicant and fits a statistical distribution to the data. From the distribution, the concentration that should theoretically protect a selected percentage of species can be determined
Toxicity	The inherent potential or capacity of a material to cause adverse effects in a living organism
Toxicity test	The means by which the toxicity of a chemical or other test material is determined. A toxicity test is used to measure the degree of response produced by exposure to a specific level of stimulus (or concentration of chemical) for a specified test period
Toxicity value	A value defining the concentration of a toxicant that represents an estimate of its toxicity to a species
TWM	Time-weighted mean
UV-Vis	Ultraviolet-visible spectrometry

Appendix A. Toxicity data that passed the screening and quality assessment and were used to derive the default guideline values

Table A.1 Summary of toxicity data that passed the screening and quality-assurance processes for total chromium (III) in freshwater

Taxonomic group (phylum)	Species	Life stage	Exposure duration (h)	Test type	Toxicity measure (test endpoint)	Test medium	Temperature (°C)	Water hardness[a] (mg/L $CaCO_3$)	pH[a]	Concentration (µg/L Cr(III))	Value used in SSD (µg/L)	Reference
Protozoa (Ciliophora)	*Spirostomum ambiguum*	NR	48	Chronic	EC50 (deformities)	Diluted Tyrod medium	25	2.8	7.2 ± 0.2	95	19[b]	Nalecz-Jawecki and Sawicki (1998)
Microalga (Chlorophyta)	*Raphidocelis subcapitata*	Exponential growth	72	Chronic	EC50 (growth, biomass)	ISO medium	24	(23)	(8.3)	17	34[b]	Vignati et al. (2010)
Microalga (Chlorophyta)	*Chlorella kessleri*	Exponential growth	72	Chronic	EC50 (growth, biomass)	ISO medium	24	(23)	(8.3)	50	10[b]	Vignati et al. (2010)
Microalga (Chlorophyta)	*Dictyosphaerium chlorelloides*	Exponential growth	72	Chronic	EC50 (growth, cell division)	BG-11 medium	20	NR	NR	2,786	557[b]	Pereira et al. (2013)
Rotifer (Rotifera)	*Brachionus calyciflorus*	Neonates	24	Acute	LC50 (mortality)	EPA medium	25	80–100	NR	1,050	–	Hernández-Ruiz et al. (2016)
Rotifer (Rotifera)	*Brachionus calyciflorus*	Neonates	24	Acute	LC50 (mortality)	EPA medium	25	80–100	NR	640	–	Hernández-Ruiz et al. (2016)
Rotifer (Rotifera)	*Lecane quadridentata*	Neonates	24	Acute	LC50 (mortality)	EPA medium	25	80–100	NR	1,279	128[d]	Hernández-Ruiz et al. (2016)
Crustacean (Arthropoda)	*Daphnia magna*	Neonates	504	Chronic	EC16 (reproduction)	Lake water	18	45	7.4–8.2	330	330	Biesinger and Christensen (1972)
Crustacean (Arthropoda)	*Daphnia similis*	Neonates	48	Acute	EC50 (immobilisation)	MS medium	20	44	6.0–6.5	3,240	324[d]	Melnikov and de Freitas (2011)
Fish (Chordata)	*Oncorhynchus mykiss*	Fertilised eggs	1,728	Chronic	NOEC (feeding fry)	NR	13	25	7.05–7.53	48	48	Stevens and Chapman (1984)

Taxonomic group (phylum)	Species	Life stage	Exposure duration (h)	Test type	Toxicity measure (test endpoint)	Test medium	Temperature (°C)	Water hardness[a] (mg/L $CaCO_3$)	pH[a]	Concentration (µg/L Cr(III))	Value used in SSD (µg/L)	Reference
Fish (Chordata)	*Poecilia reticulata*	NR	96	Acute	LC50 (mortality)	Soft water	25	20	7.5	3,330	333[d]	Pickering and Henderson (1966)
Fish (Chordata)	*Carassius auratus*	NR	96	Acute	LC50 (mortality)	Soft water	25	20	7.5	4,100	410[d]	Pickering and Henderson (1966)
Fish (Chordata)	*Pimephales promelas*	NR	96	Acute	LC50 (mortality)	Soft water	25	20	7.5	5,070	507[d]	Pickering and Henderson (1966)
Fish (Chordata)	*Lepomis macrochirus*	NR	96	Acute	LC50 (mortality)	Soft water	25	20	7.5	7,460	746[d]	Pickering and Henderson (1966)

NR = not reported.

[a] Values in parentheses were deduced from the corresponding test media protocols.

[b] Chronic EC50 converted to an estimated chronic NOEC/EC10 by dividing by 5 (Warne et al. 2018).

[c] Geometric mean

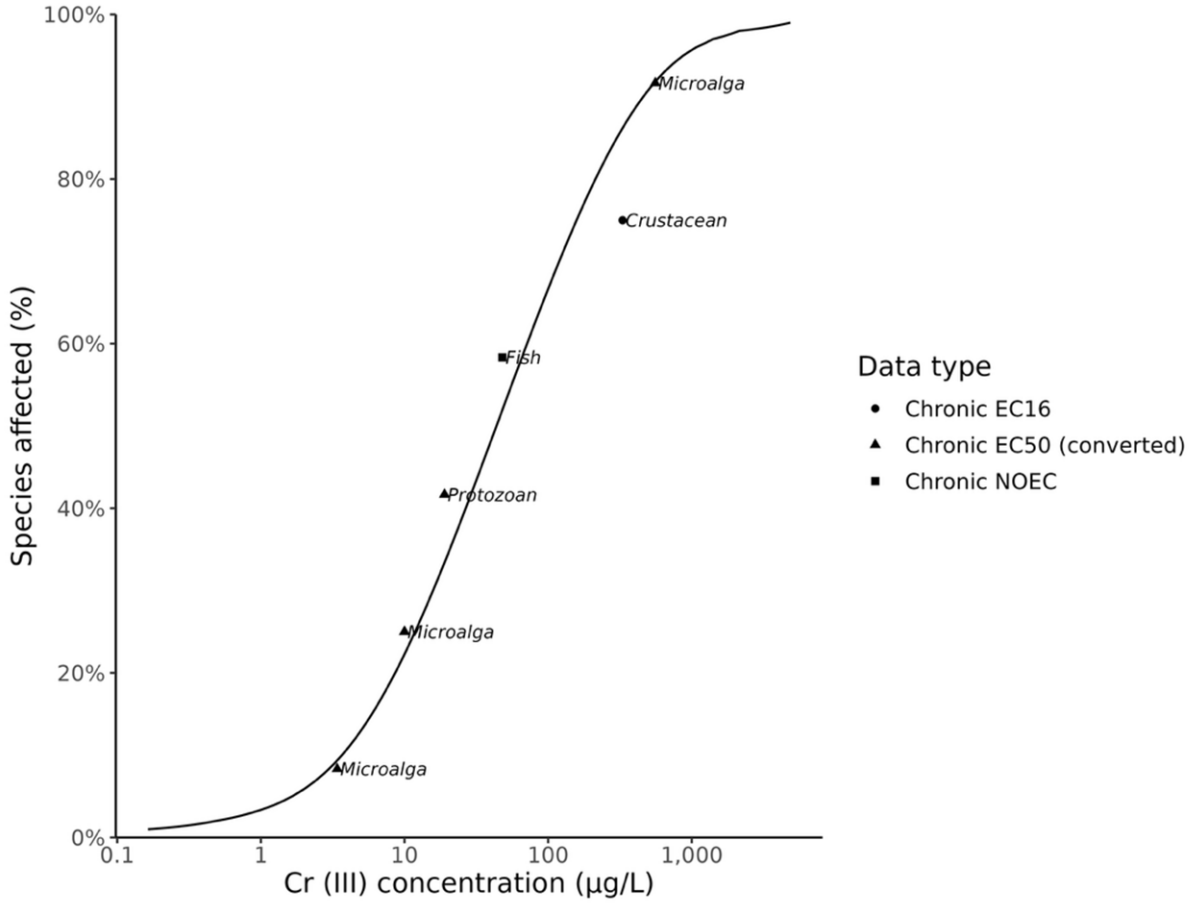
[d] Acute EC/LC50 converted to an estimated chronic NOEC/EC10 by dividing by default acute-to-chronic ratio of 10 (Warne et al. 2018).

Appendix B. Rationale for selecting the final dataset for deriving the default guideline values for total chromium (III) in freshwater

A number of iterations of the toxicity data were undertaken to attempt to improve the fit of the SSD curve and associated DGVs. This included the use of only chronic toxicity data and the use of all chronic and acute (converted to chronic) toxicity data that passed the quality-assessment and screening processes.

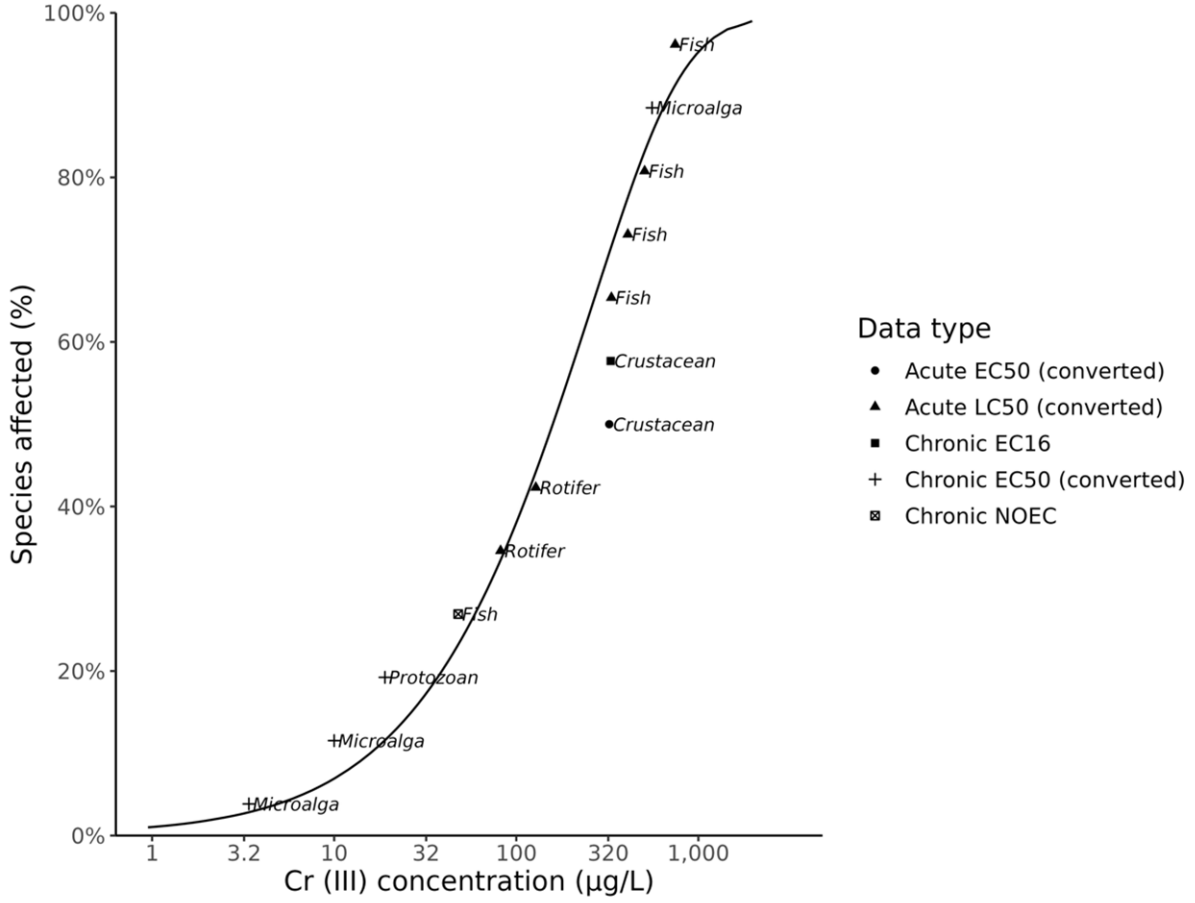
The SSD for chronic data only (6 species, 4 taxonomic groups) that passed the quality-assessment and screening processes is presented in Figure B.1. The protective concentrations (PCs) for 99%, 95%, 90% and 80% of species were 0.16 µg/L, 1.7 µg/L, 3.7 µg/L and 8.7 µg/L, respectively. The fit of the model was good, which would result in moderate reliability values according to Warne et al. (2018). However, a key limitation of this dataset was that 4 of the 6 chronic toxicity values (3 microalgae and one protozoan) were estimated negligible-effect values from reported chronic EC50 values and, therefore, carried additional uncertainty. Also, comparison of the PC values to reported natural background concentrations of chromium ((III) and (VI)) in freshwater (0.52–10 µg/L) suggested that the DGVs derived using this dataset would be overly conservative.

Figure B.1 Species sensitivity distribution for chromium (III) in freshwater for chronic toxicity data only (using the shinyssdtools [V 0.4.0] software)



The SSD for all toxicity data (chronic and acute; 13 species, 5 taxonomic groups) that passed the quality-assessment and screening processes is presented in Figure B.2. The dataset included the data presented in Table 1 and Table A.1. The PC99, PC95, PC90 and PC80 values were 0.95 µg/L, 6.7 µg/L, 16 µg/L and 39 µg/L, respectively. The fit of the model was deemed to be good, resulting in values of moderate reliability. Notably, the chronic and acute dataset resulted in slightly higher PC values compared to those for the chronic-only dataset. The main limitation of this dataset was the same as that for the chronic-only dataset, in that only 2 of the toxicity values (one fish and one crustacean) were reported chronic negligible-effect values. However, the dataset provided substantially greater taxonomic diversity than the chronic only dataset (i.e. 7 more species and one additional taxonomic group). The dataset did not appear to exhibit any characteristics of bimodality, as detailed in Warne et al. (2018). In particular, the following was noted for modality: the bimodality coefficient (BC) for the (log-transformed) dataset was 0.514, below the indicative threshold for bimodality of 0.555, there is no evidence of a specific mode of action for chromium (III) toxicity, and the dataset showed no indication of taxa-specific sensitivity. Based on this comparison of the 2 datasets, the combined chronic and acute (converted to chronic) dataset was used to derive the DGVs for total chromium (III) in freshwater (as described in Section 4).

Figure B.2 Species sensitivity distribution for chromium (III) in freshwater for all chronic and acute (converted to chronic) toxicity data that passed quality-assessment and screening processes (using the shinyssdtools [V 0.4.0] software)



Appendix C. Details of the shinyssdtools SSD-fitting process for the chromium (III) freshwater toxicity dataset

The SSD software package shinyssdtools (V 0.4.0) uses a model-averaging approach to generate an SSD (as described by Fox et al. 2021). The full set of default distributions fitted to the chromium (III) freshwater toxicity dataset is shown in Figure D.1. The relevant goodness of fit and weighting estimates for each of the distributions is shown in Table D.1. The final, model-averaged SSD is shown in Figure 1 of the main report.

Figure C.1 Species sensitivity distributions for the set of 6 default distributions used to model the chromium (III) freshwater toxicity dataset

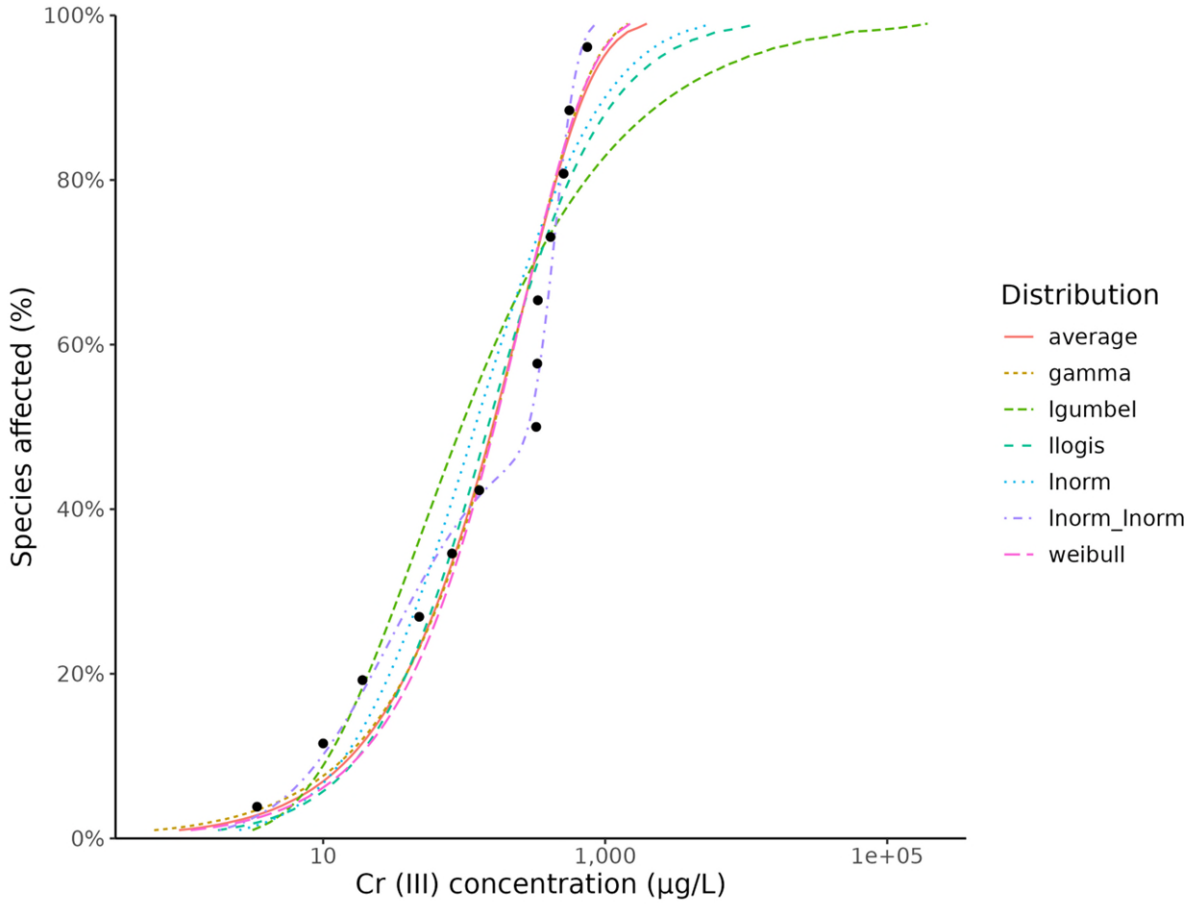


Table C.1 Goodness of fit and weighting estimates for the 6 default distributions used in model averaging for the chromium (III) freshwater toxicity dataset

Distribution	Akaike's Information Criterion	Corrected Akaike's Information Criterion	Delta	Weight
Gamma	175	176	0	0.469
Weibull	175	176	0.42	0.381
Log-normal	178	180	3.75	0.072
Log-logistic	179	180	4.27	0.055
Log-normal–log-normal	174	183	7.07	0.014
Log-gumbel	182	184	7.84	0.009

Appendix D. Water-quality assessment for sparingly soluble metals

A number of metals have low solubility in freshwater and marine water but have been found to be toxic at concentrations above their solubility limit. Therefore, the derivation of guideline values for those metals included data for which toxicity was at least in part due to particulate (precipitated) metals. Examples include iron (III) in marine water (solubility < 0.03 µg/L; Liu and Millero 2002), iron (III) in freshwater (solubility < 0.05 µg/L; Phippen et al. 2008), chromium (III) in freshwater (solubility < 5 µg/L; Rai et al. 1989) and aluminium in marine water (solubility ca 500 µg/L; Angel et al. 2016). The DGVs for these metals are expressed as total metal concentrations.

For iron and chromium, the DGVs are above the solubility limits under oxic conditions and neutral pH. Measuring total metal concentrations to compare with these DGVs requires a method that discriminates between precipitated metals and metals in mineralised forms that are not likely to be bioavailable. This is normally a cold dilute-acid (pH-2) extraction (e.g. as per US EPA 1991) that will solubilise precipitated metal oxyhydroxides, including those that become adsorbed to other substrates such as mineralised forms or particulate (or colloidal) organic matter (Markich et al. 2001). A total recoverable metals analysis (concentrated acid digestion) is not advisable, as this will overestimate the precipitated metals fraction by also including the digested mineralised forms, potentially leading to false exceedance of the DGV. At least in the case of iron and chromium, 0.45-µm sample filtration is not a recommended step, as it will exclude colloidal and precipitated metal that might be contributing to toxicity, potentially leading to false compliance of the DGV.

In a study specifically commissioned to address public comments on the chromium (III) in freshwater DGVs relating to the appropriate chemical analysis method, ANZG (2025) validated the US EPA (1991) 200.1 method for chromium (III) in both freshwater and seawater samples. The method involves extraction at pH 1.75 ± 0.1 for 16 hours and was found to provide reasonable recovery of chromium from all types of water tested. The method is applicable for measuring dissolved and precipitated or lightly adsorbed fractions of chromium in fresh and marine water. Consequently, the method described by ANZG (2025) is recommended when analysing chromium for the purpose of comparing concentrations with the chromium (III) DGVs. It is important to note that, once acidified, all of the chromium in the sample will be present as chromium (III), even if some chromium (VI) was present in the unacidified sample (ANZG 2025). Consequently, if the acid-extractable chromium concentration exceeds the chromium (III) DGV, a separate chromium (VI) analysis may be required to characterise the relative proportions of chromium (III) and chromium (VI) in the original sample. The pH-1.75 acid-extraction method is analogous to the US EPA (1991) 200.1 method, which was designed to determine acid-soluble metals but had not been validated for chromium (III). There is a low risk of toxicity if the pH-1.75 extractable fraction does not exceed the DGV, but there is potential for toxicity if the DGV is exceeded. Although it is also possible to compare the total chromium concentration of the unfiltered water sample with the DGV, if this fraction exceeds the DGV, it will not be possible to identify the relative proportions of non-bioavailable and bioavailable chromium in the sample.

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